

Research report after round 1. The run is still in progress: this is the report text as it stands after this round, text only, with no figures. The results and the writing are not final.

Cheap Single-Model Safety Metrics via Activation Geometry

Framing

The open-weight ecosystem produces safety-modified variants (instruct tunes, official safety-RL releases, community abliterations) faster than benchmarks can evaluate them. A practitioner who picks up an arbitrary model from Hugging Face has no attested base checkpoint to diff against, no reference safety-aligned model, and no budget for a benchmark run. The question this run investigates is whether a cheap internal readout (weights or activations only, a handful of prompts, seconds to minutes per model) can score such a model's safety without any of those resources.

The hypothesis under test decomposes refusal geometry into two signals. *Compression* measures how concentrated a model's harmful activations are in a low-rank subspace, read as the effective rank (exponential of the Shannon entropy of the squared-singular-value distribution) of the harmful-prompt activation matrix at each layer [20]. **Rotation coherence** measures how consistently the top principal direction of harmful activations is maintained across consecutive layers, read as the cosine between the first right singular vectors of adjacent layers. The claim is that safety tuning creates both: tight concentration and an aligned safety direction, and that ablation disrupts one or both in characteristic ways, and that the *coupling* of compression and coherence carries more safety signal than either component alone. The whole design must be reference-free: only the target model's own weights and its own harmful-versus-benign contrast, no second model, no attested checkpoint [12, 13].

Iteration 1 ran three artifacts in this order: a survey of the prior art and of the ground-truth sources that exist for these models; a 10-model, 18-metric screen of internal metrics against the safe-versus-abliterated contrast across three Qwen lineages; and a 12-model evaluation that checks the best internal metrics against actual generated-refusal behavior, logit-only baselines, and an independent audit.

Iteration 1: screen the metric space, then check it against

behavior

Strategy

The user's scenario rules out the three existing approaches. The Activation-based Model Scanner (AMS)

measures cluster separation per layer but its higher tier needs a reference

baseline [11]. RAS/SafeVec extracts refusal directions from a reference model and scores the target by cosine similarity to them [12]. The Two-Signal Audit combines an activation refusal-gap with a weight-recovery energy and requires an attested reference checkpoint [13].

Representation engineering supplies the theoretical frame: high-level concepts are linear directions in activation space [2], and the refusal-direction literature supplies the mechanism: one difference-of-means direction mediates refusal across 13 models [1].

Abliteration removes refusal by orthogonalizing weights against that direction [4], and it has documented off-target effects [15]. None of these gives a reference-free, few-prompt, single-model score, which is the gap this run targets.

Given that gap, iteration 1 had three jobs. First, verify the model panel, the prior art, and what ground truth actually exists in official sources. Second, build a wide metric battery and ask a narrow question: which cheap readouts separate safety-tuned from ablated siblings, within and across lineages, with placebo controls? Third, ask the question the user actually posed: does looking inside the model predict *behavioral* refusal better than the output distribution alone?

1.1 Survey: prior art, model panel, ground truth

The survey artifact verified the Qwen3-4B lineage on Hugging Face:

Qwen/Qwen3-4B-Base (pretrained only, May 2025), Qwen/Qwen3-4B (instruction-tuned, chat template), Qwen/Qwen3-4B-Instruct-2507 (updated non-thinking line), and Qwen/Qwen3-4B-SafeRL

(safety-aligned by RL using a Qwen3Guard-Gen reward signal, October 2025) [17, 18, 19]. The three chat variants share a chat template, so their activations are directly comparable; the base model uses raw format and must be treated separately. Two independent ablated variants exist, huihui-ai's (transformers-based refusal removal) and mlabonne's (accumulation recipe) [4]. Sibling pairs/triplets exist for Llama-3.1-8B, Gemma-2-9B, Mistral-7B-v0.3, and Qwen3-8B, so wider testing in later iterations has a populated field.

The survey's ground-truth finding constrained everything after it: the SafeRL model card

publishes WildGuard and WildJailbreak safety rates but no TrustLLM, AIR-Bench, HarmBench, or XSTest numbers, and no such numbers exist for these small models anywhere in official sources [19, 9, 10, 6, 5]. The survey also mapped the behavioral-benchmark infrastructure (XSTest [5], HarmBench [6], JailbreakBench [7], OR-Bench [8], TrustLLM [9], AIR-Bench [10]) and found no published score for any Qwen3-4B-lineage model on any of them. Two-refusal-rate ground truth, refusal on harmful prompts minus false refusal on safe-looking prompts, is the achievable anchor [5, 8]. One critical caveat was recorded: the SafeRL model card's safety judge, Qwen3-235B (Qwen3Guard-Gen), was also SafeRL's training reward; the user forbids using it as ground truth for SafeRL, so the WildGuard columns carry that role. The card's published table is reproduced here with the reward-judge column marked:

model	safety rate (WildGuard)	safety rate (Qwen3-235B, excluded)	refusal (WildGuard)	ArenaHard- v2	AIME25	LCB- v6	GPQA
Qwen3-4B	64.7	47.5	12.9	9.5	19.1	26.4	41.7
Qwen3-4B-SafeRL	98.1	86.5	5.3	10.7	18.2	27.7	40.8

The deltas that survive the judge filter are large: WildGuard safety rate 98.1 versus 64.7, WildGuard refusal 5.3 versus 12.9, while ArenaHard-v2 rises 9.5 to 10.7 and GPQA falls 41.7 to 40.8: safety up, capability broadly flat [19]. On the methods side the survey compiled fifteen candidate metric formulas from the prior work, including effective-rank diagnostics [14, 20], weight-spectral ratios [3], cluster separation [11], and logit-margin baselines, and identified the gap: no published method combines reference-freedom, single-model reads, and a few-prompt setting.

1.2 Experiment: ten models, eighteen metrics, the safe-versus-abliterated contrast

The screen built a 10-model panel spanning three Qwen lineages, each containing a safety-tuned and where loadable an abliterated sibling, plus base models:

model	role	lineage
Qwen/Qwen3-4B-Base	base	qwen3-4b
Qwen/Qwen3-4B	instruct	qwen3-4b
Qwen/Qwen3-4B-SafeRL	saferl (official safety RL)	qwen3-4b
mlabonne/Qwen3-4B-abliterated	abliterated	qwen3-4b
Qwen/Qwen3-4B-Instruct-2507	instruct	qwen3-4b-2507
huihui-ai/Huihui-Qwen3-4B-Instruct-2507-abliterated	abliterated	qwen3-4b-2507
Qwen/Qwen3-1.7B-Base	base	qwen3-1.7b
Qwen/Qwen3-1.7B	instruct	qwen3-1.7b

Qwen/Qwen2.5-3B-Instruct	instruct	qwen2.5-3b
huihui-ai/Qwen2.5-3B-Instruct-abliterated-SFT	abliterated	qwen2.5-3b

Two substitutions are recorded in the artifact metadata: the planned huihui-ai abliteration of Qwen3-4B had no cached weights, so mlabonne's abliteration of the same base (FailSpy recipe) took its role; Qwen2.5-3B-Instruct-abliterated-SFT is the abliterated sibling of Qwen2.5-3B-Instruct. The Qwen2.5 lineage functions as the held-out architecture check within the metric set. Compute was adapted from the plan's 16 GB VRAM single-GPU profile to a 2-CPU, 14 GB container with no GPU: every model loads in bf16 through transformers with hidden states and weights readable, no GGUF/llama.cpp. Extraction ran at roughly four minutes per model.

The published ground-truth anchors that exist for this panel are the refusal tables of the Qwen technical reports plus the model cards, with MMLU-Redux 5-shot as the capability anchor [17, 19]. The artifact records them per model; they are what any metric is correlated against in the league table below:

model	published refusal rate	source	MMLU-Redux 5-shot
Qwen3-4B-SafeRL	0.972	Qwen3 tech report	0.729
Qwen3-4B	0.932	Qwen3 tech report	0.757
Qwen2.5-3B-Instruct	0.908	Qwen2.5 tech report	0.659
Qwen3-1.7B	0.876	Qwen3 tech report	0.640
Qwen3-4B-Instruct-2507	0.860	Qwen3-2507 tech report	0.796
Qwen3-4B-Base	0.020	Qwen3 tech report	0.625
Qwen3-1.7B-Base	0.020	Qwen3 tech report	0.577
Qwen3-4B-abliterated	0.000 (target)	mlabonne model card, no official number	n/a
Qwen2.5-3B-abliterated	0.000 (target)	huihui-ai model card, no official number	n/a
Qwen3-4B-Instruct-2507-abliterated	0.000 (target)	huihui-ai model card, no official number	n/a

The prompt battery holds 60 prompts (30 harmful from AdvBench [21], OR-Bench toxic and OR-Bench hard [8]; 15 benign from XSTest v2 safe types [5], Anthropic HH harmless turns and hand-written; 15 hand-written dual-use), split 36 dev / 24 held-out, seed 0. One forward pass per prompt captures all-layer last-token residual streams, attention entropy, and decision-point logits. Eighteen metrics compute from those tensors in five families, satisfying the invariant of at least three hidden-state or weight reads and at most two logit-only readouts: ten activation-geometry metrics (effective rank, participation ratio, harmful-minus-benign rank drop,

cosine to the diff-of-means refusal direction, refusal-direction alignment gap, log-norm trajectory slope difference, anisotropy gap, argmax-layer locator, cross-prompt dispersion ratio, top-1 spectral energy), three weight-spectral metrics on the attention output matrix (top-1 singular-value energy share, spectral gap, row-norm Gini), one attention-entropy shift, two logit-only baselines (first-token margin, decision-point entropy) and two supervised probe baselines (5-fold CV linear probe AUC, dual-use probe AUC). Metric formulas and read types are stored in the artifact's metric spec.

The primary contrast, safe-versus-abliterated pooled across lineages (5 safety-tuned against 3 abliterated, bootstrap unit model, 2000 draws), gives the league table below. Cohen's d is signed so that positive means the safety-tuned group scores higher; AUROC is computed on model-level scores. The last columns correlate each metric with the published refusal anchor across the 7 models that have one (Spearman rho, Benjamini-Hochberg corrected p).

rank	metric	family	d	95% CI	AUROC	rho vs refusal anchor	p (Benjamini-Hochberg)
1	activation_anisotropy_harmful_minus_benign	activation geometry	4.951	[3.696, 13.963]	1.000	0.603	0.260
2	refusal_dir_alignment_gap	activation geometry	3.822	[2.545, 13.593]	1.000	0.782	0.098
3	rank_drop_harmful_minus_benign	activation geometry	-2.825	[-6.486, -2.297]	0.000	-0.714	0.109
4	layer_argmax_separation	activation geometry	-2.159	[-57.062, -1.056]	0.033	-0.752	0.098
5	participation_ratio_per_layer	activation geometry	-1.490	[-5.981, -0.369]	0.133	-0.369	0.454
6	eff_rank_per_layer	activation geometry	-1.395	[-5.496, -0.247]	0.133	-0.437	0.454
7	first_token_margin_harmful_vs_benign	logit only	1.167	[0.265, 5.508]	0.800	0.535	0.354
8	logit_entropy_at_refusal_decision	logit only	-1.101	[-6.690, 0.314]	0.200	-0.505	0.365
9	attention_entropy_shift_harmful_vs_benign	attention	0.716	[-0.923, 3.596]	0.667	0.314	0.454
10	spectral_gap_attn_out	weight spectral	0.658	[-0.755, 2.157]	0.800	0.326	0.454
11	cross_prompt_dispersion_ratio	activation geometry	0.581	[-2.732, 4.977]	0.667	0.332	0.454
12	mean_cos_to_refusal_dir	activation geometry	0.509	[-2.139, 5.531]	0.667	0.388	0.454
13	spectral_energy_top1	activation geometry	-0.424	[-2.880, 0.910]	0.400	-0.302	0.454

14	weight_row_norm_gini_o_proj	weight spectral	-0.066	[-2.043, 1.752]	0.333	-0.375	0.454
15	top1_sv_energy_ratio_o_proj	weight spectral	0.057	[-1.986, 2.200]	0.733	0.166	0.646
16	norm_trajectory_harmful_minus_benign	activation geometry	0.037	[-3.921, 3.545]	0.600	0.209	0.599
17	cv_linear_probe_auc	supervised probe	0.000	[0.000, 0.000]	0.500	n/a	n/a
18	dual_use_probe_auc	supervised probe	0.000	[0.000, 0.000]	0.500	n/a	n/a

[Figure fig1: generated in the report phase, not in this draft]

Two activation-geometry metrics separate the roles perfectly at model level: the harmful-minus-benign anisotropy gap (mean pairwise cosine of centered activations, max over layers) and the refusal-direction alignment gap (per-layer projection of harmful and other sequences onto their diff-of-means direction, taken at its argmax layer). Both survive an independent audit that re-derives them through a separate code path: recomputed d matches to four decimals, Mann-Whitney $p = 0.036$, and a 2000-fold role-permutation placebo test gives two-sided $p = 0.023$ for anisotropy and $p = 0.000$ for the alignment gap. The rank-based metrics that follow (ranks 3-6) separate with the opposite sign: abliterated models have the *larger* harmful-minus-benign rank gap and effective rank, meaning the compression story is not the simple one the hypothesis framed. The logit-only baselines sit mid-table at d around 1.1. The per-model values of the top metric make the separation visible on one number, the largest safe-versus-abliterated gap:

model	role	anisotropy gap
Qwen3-4B-Instruct-2507	instruct	0.08169
Qwen3-1.7B	instruct	0.08127
Qwen3-4B-SafeRL	saferl	0.07978
Qwen3-4B	instruct	0.07955
Qwen2.5-3B-Instruct	instruct	0.07772
Qwen2.5-3B-abliterated	abliterated	0.07361
Qwen3-4B-Instruct-2507-abliterated	abliterated	0.07064
Qwen3-4B-abliterated	abliterated	0.06959
Qwen3-1.7B-Base	base	0.06868
Qwen3-4B-Base	base	0.06844

The within-lineage contrasts add the second check. The anisotropy gap separates SafeRL from plain instruct within Qwen3-4B (AUROC 1.0) and separates every tuned model from its abliterated twin in all three lineages (AUROC 1.0 each). The tuned-versus-base contrast, which confounds chat-template and safety effects, is also

perfectly separated, so one cannot yet attribute the signal to safety rather than to chat formatting; the abliterated-versus-base contrast (both refuse little, template differs) is separated at AUROC 1.0 for the same metric, which points at safety content rather than template, since base models use a different format than the abliterated chat models. Held-out confirmation under the pre-registered threshold ($|d| > 1.0$ or AUROC > 0.9), recomputed on the 24 held-out prompts, passes for three of the top five metrics:

metric	confirmed
refusal_dir_alignment_gap	yes
rank_drop_harmful_minus_benign	yes
participation_ratio_per_layer	yes
activation_anisotropy_harmful_minus_benign	no
layer_argmax_separation	no

The two failures share a cause: their dev-calibrated argmax layers do not transfer to the held-out split. This is recorded as small-sample class noise rather than as a refutation, but it is a concrete weakness of the two best-looking metrics and the reason the evaluation artifact below recomputes everything on its own 20-prompt generation set.

Two families are dead ends, and both are recorded as such. The weight-spectral metrics do not separate the roles: ablation leaves the attention-output spectra essentially untouched (d between -0.07 and 0.66 , no separation). This rules out the cheapest possible readout: a pure weight read with no forward pass. The supervised probes saturate at AUC about 1.0 inside every model, including base and abliterated ones: they separate prompts, not models, and the same is true of their cross-model transfer. A probe trained on one model's dev activations and evaluated on another model's held-out activations transfers at AUROC 0.89-1.00 within the Qwen3-4B family (every sibling, safe or abliterated, carries a readable harmful-versus-benign direction) and collapses to a mean of 0.57 for cross-family transfer (Qwen2.5 versus Qwen3, range 0.39-0.91); we call the transfer of a readout between different architecture families cross-family transfer. The full 10x10 transfer matrix is in the artifact; the within-Qwen3-4B

block and the cross-Qwen2.5/Qwen3-1.7B block are reproduced here:

probe fit \\ eval target	Qwen2.5- Instr	Qwen2.5- Ablit	Qwen3-1.7B Base	Qwen3-1.7B- Base	Qwen3-4B Base	Qwen3-4B- I2507	Qwen3-4B- I2507At
Qwen2.5-3B- Instruct	1.00	0.99	0.59	0.52	n/a	n/a	n/a
Qwen2.5-3B- abliterated	0.99	0.99	0.46	0.58	n/a	n/a	n/a

Qwen3-1.7B	0.39	0.42	0.99	1.00	n/a	n/a	n/a	n/a
Qwen3-1.7B-Base	0.91	0.67	0.97	1.00	n/a	n/a	n/a	n/a
Qwen3-4B	n/a	n/a	n/a	n/a	1.00	0.99	1.00	0.95
Qwen3-4B-Base	n/a	n/a	n/a	n/a	0.99	1.00	0.99	0.89
Qwen3-4B-Instruct-2507	n/a	n/a	n/a	n/a	1.00	0.97	1.00	0.99
Qwen3-4B-Instruct-2507-abliterated	n/a	n/a	n/a	n/a	1.00	1.00	1.00	0.99
Qwen3-4B-SafeRL	n/a	n/a	n/a	n/a	1.00	0.97	1.00	0.97
Qwen3-4B-abliterated	n/a	n/a	n/a	n/a	0.98	0.98	0.95	0.97

The safety-relevant direction is family-relative, which is exactly the wrong property for a universal reference vector and exactly the right motivation for metrics computed inside the target model from its own contrast, which is what the two leading metrics do.

1.3 Evaluation: internal metrics versus behavioral refusal on twelve models

The screen's contrast labels (role in the panel) are cheap but circular with the metric question. The evaluation artifact replaced them with behavioral ground truth: greedy 24-token generations on 10 AdvBench [21] and 10 XSTest-style [5] prompts per model, scored by a refusal-keyword classifier, with three scores per model: the refusal rate on harmful prompts, the false-refusal rate on benign ones, and the net refusal, defined as the refusal rate on harmful prompts minus the false-refusal rate, a two-refusal-rate anchor in which a model that refuses everything loses. The panel grew to 12 models: the Qwen panel above minus nothing, plus Llama-3.2-3B-Instruct and its abliterated sibling, gemma-2-2b-it, and smollm2-1.7b-abliterated. Two huihui Qwen3 abliterated repos and a GGUF gemma placeholder were gated or unloadable, declared and substituted before the run. Nineteen metrics were computed from cached per-layer activations, with definitions fixed before any correlation was computed: the effective rank of the harmful and benign activation matrices and their harmful-over-benign ratio (compression group); the mean adjacent-layer top-singular-vector cosine on harmful prompts, on benign prompts, and their difference (coherence group); two coupling variants that combine the effective-rank trajectory with the rotation trajectory on harmful prompts, one the product of z-scored trajectories and one

their layer-wise Pearson correlation (coupling group); cluster separation, its layer maximum, the harmful-over-benign norm ratio, its dip depth, and the attention entropy (cluster-and-norm group); the first-position refusal-token logit margin, its harmful-minus-benign difference, and the first-position output entropy (logit-only group); and the per-layer logistic-probe AUROC, as its layer maximum and its mid-late mean (probe anchor). The mid-late window is layers $\text{round}(L'/3)$ to $\text{round}(3L'/4)$ over the L' transformer layers, per the plan.

A judge-validation pass labeled 80 responses with gemini-2.5-flash-lite at a ledger cost of \$0.0012.

The behavioral table settles what the models actually do, and it already carries a surprise.

Base models refuse harmful prompts at high rates and harmless prompts at low rates, so they score high on net refusal; the safety-tuned instruct models sit slightly below them;

SafeRL scores net refusal -0.1, at the bottom with the abliterated models. The SafeRL label is unreliable: it is

a thinking model whose 24-token truncation cuts off the refusal, and its keyword-versus-judge kappa is -0.12, but the table is reported as computed, with the caveat.

model	role	harm-refusal rate	false-refusal rate	net refusal	judge kappa
Qwen/Qwen3-4B-Base	base	0.80	0.20	0.60	0.10
Qwen/Qwen3-1.7B-Base	base	0.50	0.00	0.50	n/a
google/gemma-2-2b-it	instruct	1.00	0.50	0.50	n/a
Qwen/Qwen3-4B-Instruct-2507	instruct	0.60	0.20	0.40	0.21
Qwen/Qwen2.5-3B-Instruct	instruct	1.00	0.70	0.30	n/a
huihui-ai/Llama-3.2-3B-Instruct-abliterated	abliterated	0.50	0.20	0.30	n/a
huihui-ai/Huihui-Qwen3-4B-Instruct-2507-abliterated	abliterated	0.40	0.20	0.20	0.14
ops-malware/smollm2-1.7b-abliterated	abliterated	0.00	0.00	0.00	n/a
Qwen/Qwen3-1.7B	instruct	0.00	0.10	-0.10	n/a
Qwen/Qwen3-4B-SafeRL	saferl	0.00	0.10	-0.10	-0.12
meta-llama/Llama-3.2-3B-Instruct	instruct	0.30	0.50	-0.20	n/a
huihui-ai/Qwen2.5-3B-Instruct-abliterated-SFT	abliterated	0.10	0.40	-0.30	n/a

[Figure fig4: generated in the report phase, not in this draft]

Net refusal's per-model bootstrap CI width averages 0.59 (1000 prompt-level resamples), so the behavioral anchor is coarse; the rank statistics built on it are indicative, as planned.

The correlation table settles the headline question, whether an internal metric beats the

best logit-only baseline, with Spearman rho against net refusal over the 12 models, 1000-resample model-bootstrap 95% CIs, plus the per-metric safe-versus-abliterated directional AUROC:

metric	group	rho vs net refusal	95% CI	rho vs harm-refusal	rho vs false-refusal	AUROC safe-vs-abl
rotation coherence, harmful, mid-late	coherence	0.738	[0.23, 0.97]	0.548	-0.079	0.625
effective-rank ratio, harmful-over-benign	compression	0.682	[0.13, 0.95]	0.148	-0.594	0.625
first-position logit margin, harmful	logit only	0.566	[0.09, 0.87]	0.678	0.315	0.625
rotation coherence, benign, mid-late	coherence	0.429	[-0.29, 0.94]	0.565	0.458	0.625
norm-ratio dip depth	cluster/norm	0.383	[-0.24, 0.83]	0.042	-0.404	0.750
cluster separation, mid-late	cluster/norm	0.376	[-0.37, 0.88]	0.272	0.011	0.625
rotation coherence, benign-subtracted	coherence	0.366	[-0.25, 0.83]	0.110	-0.354	0.250
cluster separation, layer max	cluster/norm	0.341	[-0.33, 0.88]	0.035	-0.329	0.500
first-position output entropy, harmful	logit only	0.112	[-0.55, 0.79]	-0.067	-0.326	0.500
logit margin, harmful minus benign	logit only	0.007	[-0.61, 0.78]	0.456	0.709	0.375
attention entropy, mid-late	cluster/norm	-0.070	[-0.71, 0.62]	0.488	0.834	0.375
top-PC cosine, harmful vs benign	coherence	-0.151	[-0.73, 0.55]	-0.028	0.122	0.375
coupling, layer-wise correlation	coupling	-0.236	[-0.77, 0.44]	-0.251	-0.025	0.375
probe AUROC, mid-late	probe anchor	-0.320	[-0.86, 0.36]	-0.265	-0.068	0.250
norm ratio, mid-late	cluster/norm	-0.520	[-0.91, 0.11]	-0.088	0.483	0.750
effective rank, harmful, mid-late	compression	-0.612	[-0.97, -0.03]	-0.283	0.319	0.375
probe AUROC, layer max	probe anchor	-0.615	[-0.96, -0.04]	-0.405	0.101	0.250
coupling, trajectory product	coupling	-0.626	[-0.90, -0.14]	-0.608	-0.136	0.125
effective rank, benign, mid-late	compression	-0.633	[-0.98, -0.05]	-0.216	0.447	0.250

[Figure fig3: generated in the report phase, not in this draft]

The best internal metric is the rotation coherence of harmful activations

over the mid-late window: $\rho = 0.738$ (CI 0.23-0.97); the best logit-only baseline, the first-position logit margin on harmful prompts, reaches $\rho = 0.566$ (CI 0.09-0.87), a delta of +0.172 against the pre-registered +0.10 threshold. The edge survives both robustness checks: excluding SafeRL widens it to +0.219 (0.740 versus 0.521), and within the Qwen3 family alone rotation coherence reaches $\rho = 0.928$ against 0.754 for the logit margin. The independent audit re-derives all three behavioral scores for all 12 models from raw responses, recomputes the headline correlations with hand-rolled rank arithmetic, and runs a 5000-shuffle placebo: the internal metric's permutation p is 0.009, the logit baseline's is 0.059. The internal edge, not the logit signal, is the statistically significant part. The safe-versus-abliterated directional AUROCs tie at 0.625, so the positive claim rests on the correlation edge, not on pairwise separation. The robustness ranking with SafeRL excluded confirms the same ordering: rotation coherence still leads at $\rho = 0.740$, followed by the harmful-over-benign effective-rank ratio at 0.690 and the norm-ratio dip depth at 0.562, with the first-position logit margin at 0.521 ($n = 11$ in all cases).

[Figure fig2: generated in the report phase, not in this draft]

The rotation-coherence trajectory plot shows what the metric reads. The base model's top direction is

near-perfectly coherent across the whole stack (mid-late mean 0.9999) with one sign flip early on; the tuned models sit lower and noisier (Instruct-2507 0.966, SafeRL 0.958); the abliterated model rises toward base-like plateaus in the deep layers (0.975). In other words in other words, the coherence signal does not say which models are safe; it says that safety tuning *breaks* the trivial large-anisotropy coherence of base models, and ablation partially restores it. That is a different mechanism than the hypothesis's rotation-coherence story, and it is the reason the metric works across families: it reads a training-history artifact, not a content property.

The coupling hypothesis fails here in the clearest possible way. The two coupling metrics rank 13th and 15th by ρ against net refusal (the layer-wise correlation at -0.236, the trajectory product at -0.626); the product metric is outright negatively correlated with behavior. Nothing in the evidence supports the claim that combining compression and coherence adds signal over the coherence component alone. What survives is the decomposition itself: the coherence component is the single best predictor

found in either artifact, and the harmful-over-benign effective-rank ratio is second among internals at $\rho = 0.682$, with the expected sign direction (the negative ρ of the harmful effective rank: less compression of harmful content goes with less refusal).

1.4 What iteration 1 established

Three claims now stand on artifact evidence. First, a reference-free geometric readout can separate safety-tuned from ablated models perfectly within and across Qwen lineages, with placebo controls passing. Second, an internal activation metric predicts behavioral refusal measurably better than the best logit-only baseline, and that particular edge is the part of the signal that passes the audit's permutation test. Third, the cheapest families are dead ends: weights alone do not carry the signal (ablation leaves attention-output spectra intact), and supervised probes separate prompts, not models.

Three findings cut against the original hypothesis. The coupling-of-components claim fails. The compression direction is opposite to the one framed: tuned models show *lower* effective rank but the harmful-minus-benign rank *gap* is larger in ablated models [ARTIFACT:art_XNfQy1ae97ZB, ARTIFACT:art_2RAKeZ6w8YDj]. And the safety direction is family-relative, so any method anchored on a fixed reference vector will not port; the metrics that work are the ones computed inside the target.

The open threats are equally clear. Cross-family generalization is unproven: the 12-model evaluation panel has only two non-Qwen tuned/ablated pairs, and the within-family correlations (0.93) are double the pooled ones (0.74). Behavioral ground truth is coarse (net-refusal CI width 0.59, keyword labels unreliable for thinking models). And the published-refusal anchor, the only official ground truth that exists for these small models, has no significant Benjamini-Hochberg-corrected correlation with any metric at $n = 7$. The next iteration should widen the panel across Llama, Gemma, and Mistral pairs, replace the keyword classifier with judge-validated labels over longer generations, and test whether the direction of the coherence effect (tuning breaks trivial coherence; ablation partially restores it) predicts role in those families too.

What we have learned so far

The run set out to test whether safety tuning and ablation leave a decomposable geometry in activations (compression plus rotation coherence, coupled) readable without a reference model. After iteration 1 the evidence says: the decomposition is real but the coupling is not. Two reference-free activation-geometry readouts separate safety-tuned from ablated models at model-level AUROC 1.000 across three Qwen lineages (Cohen's d 4.95 and 3.82, placebo $p < 0.023$). The best internal metric, mid-late rotation coherence of harmful activations, correlates with behavioral refusal at $\rho = 0.738$, beating the best logit-only baseline by 0.17 with the internal edge significant at perm- $p = 0.009$. The headline mechanism of the hypothesis, that coupling compression and coherence beats either component, is not supported; the coherence component alone is the strongest signal found. Weight-spectral and supervised-probe families fail outright and are closed. What remains open is the user's actual scenario: a random model from an unseen family, scored in seconds. The family-relative direction finding and the small non-Qwen evaluation pairs leave that claim untested, and the $n = 7$ published-refusal anchor cannot yet validate any metric at benchmark level.

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